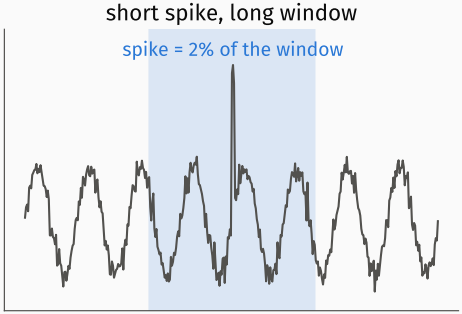
Looking at a time series at many scales

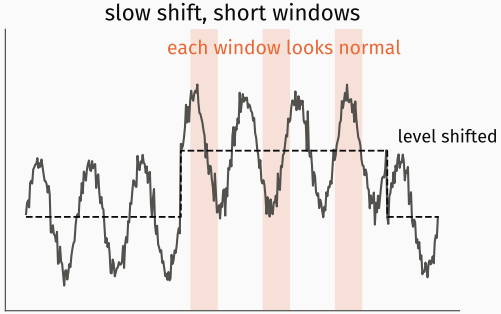
The Haar ladder: theory, implementation, results (stage 0)

DG_AD

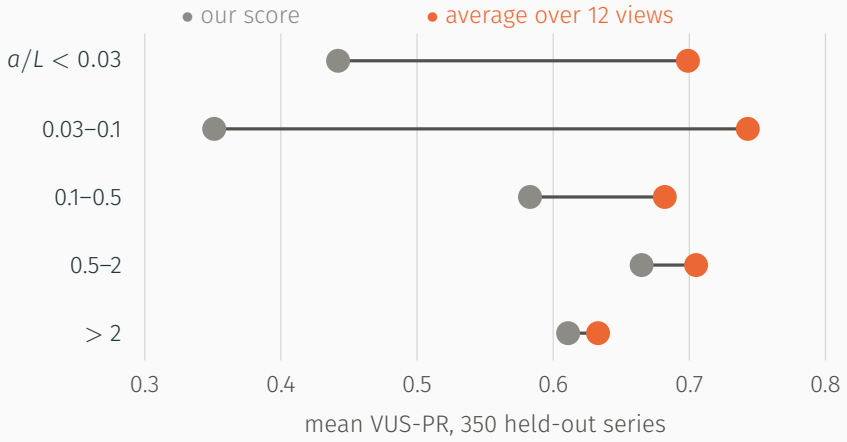
23 October 2026

One window, two failures



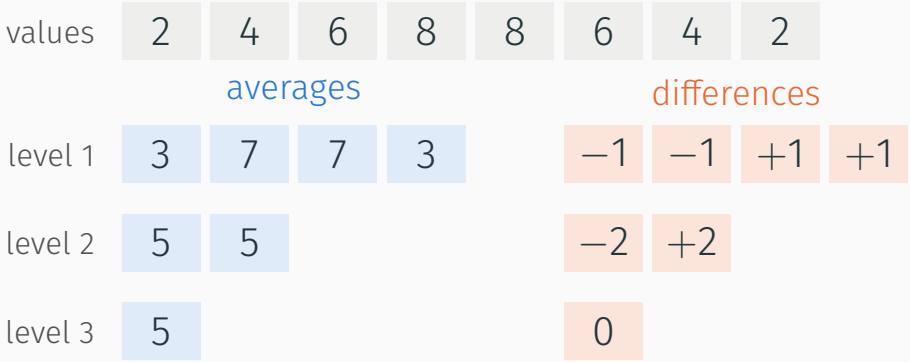


Where we lose: short anomalies



Theory

Averages and differences



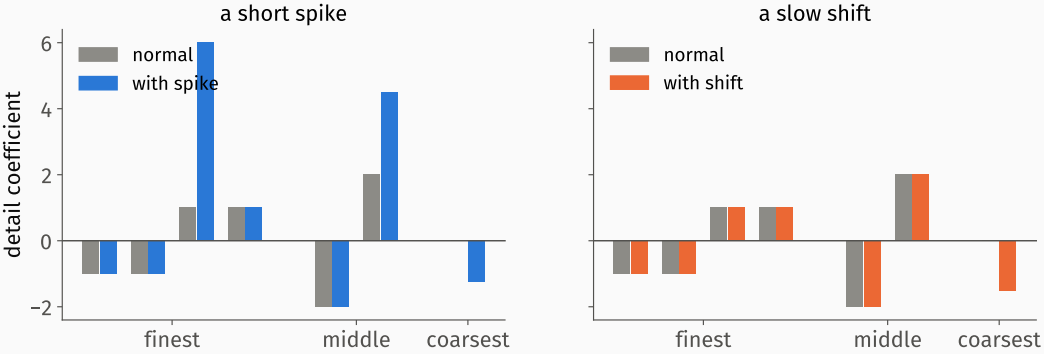
Nothing is lost

$$x_1 = a + d \qquad x_2 = a - d$$

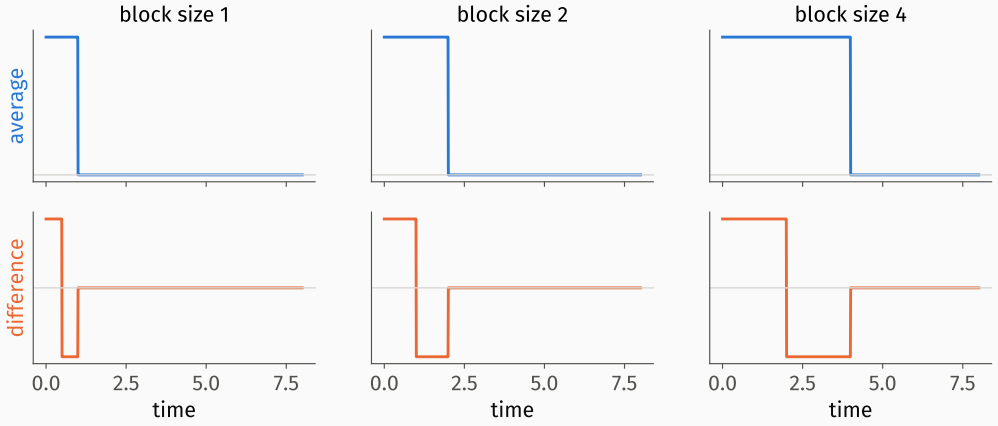
8 values $\longleftrightarrow$ 1 average + 7 differences

the same signal, arranged by scale

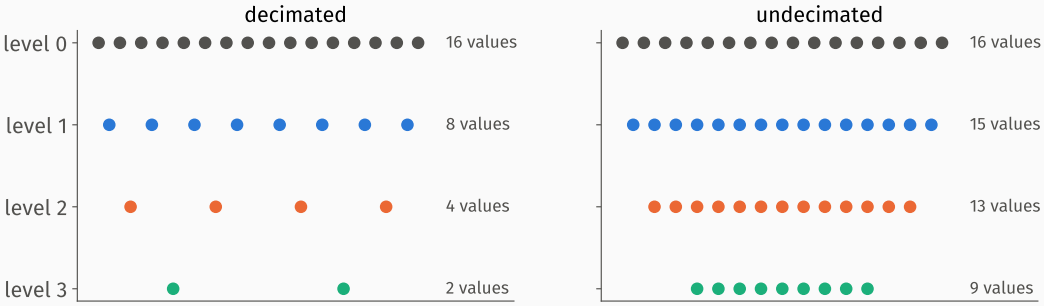
Spikes live at fine scales, shifts at coarse ones



The Haar wavelet

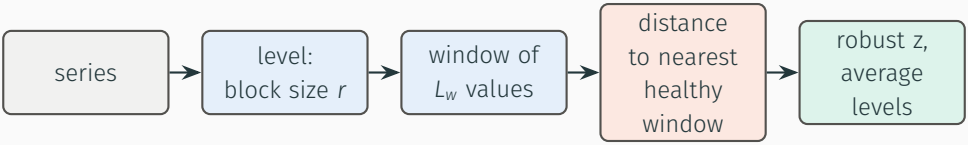


Decimated or undecimated

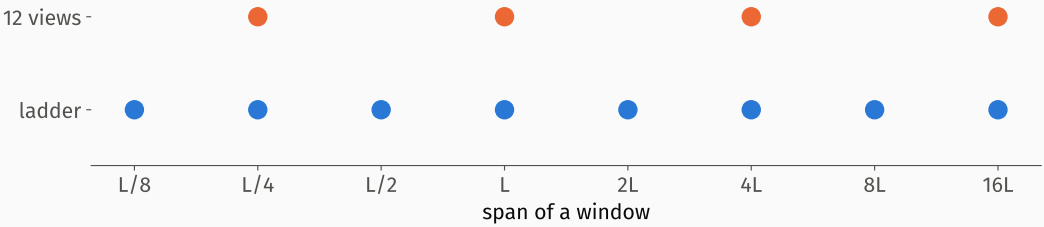


Implementation

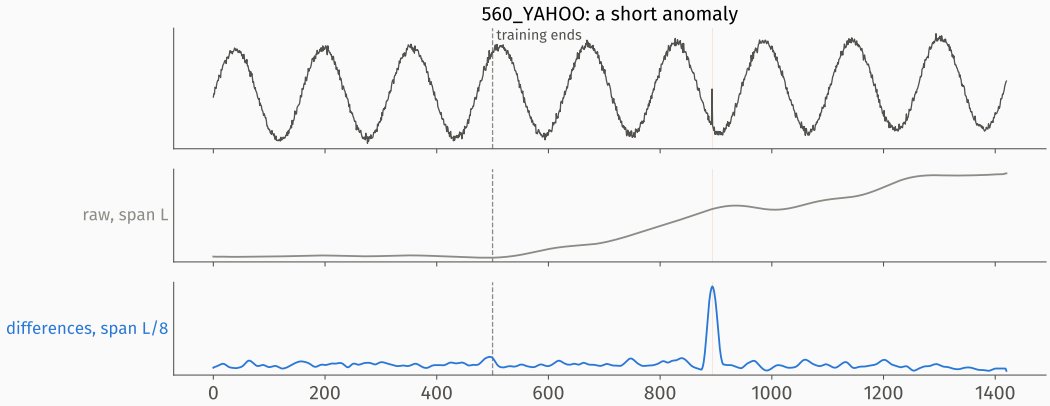
From a level to a score



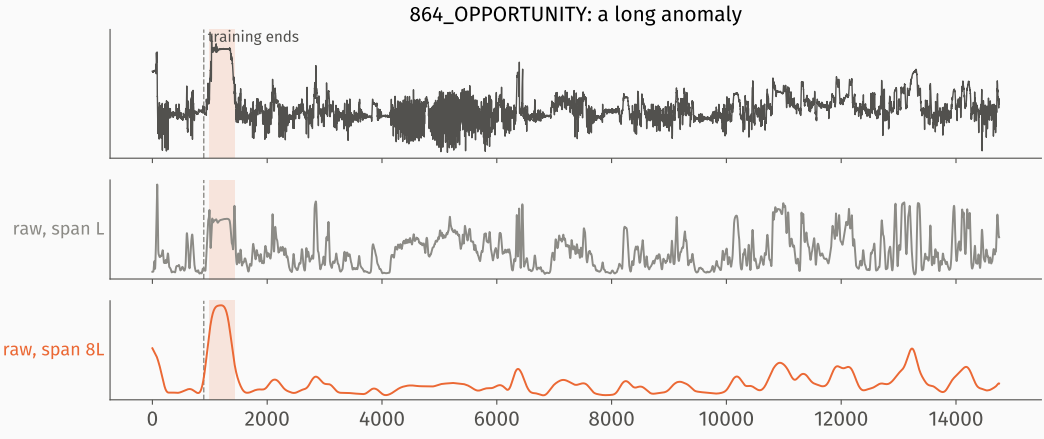
Our 12 views were a partial ladder



A fine scale catches a short anomaly

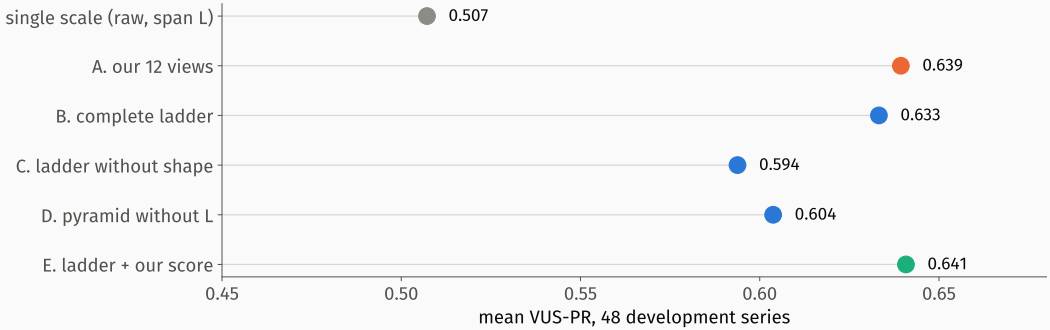


A coarse scale catches a long anomaly



Results

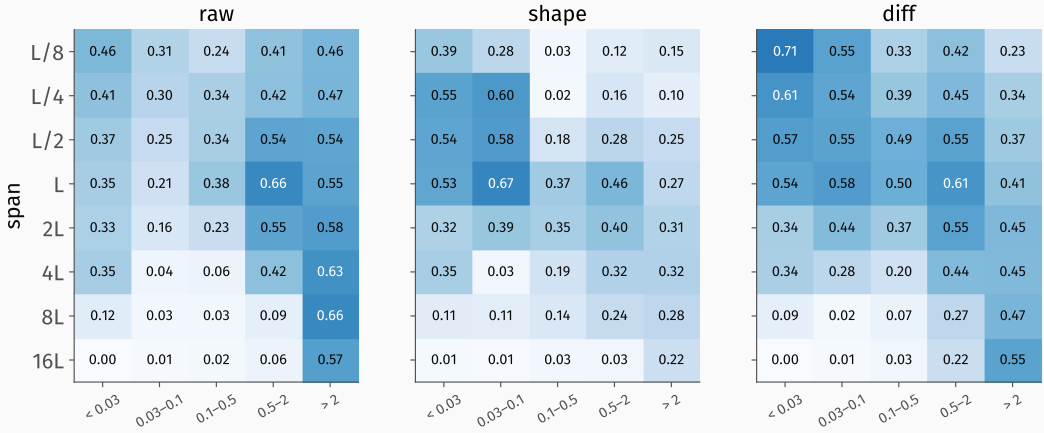
Five arms


single scale (raw, span L)
A. our 12 views
B. complete ladder
C. ladder without shape
D. pyramid without L
E. ladder + our score
0.507
0.639
0.633
0.594
0.604
0.641
0.45
0.50
0.55
0.60
0.65
mean VUS-PR, 48 development series

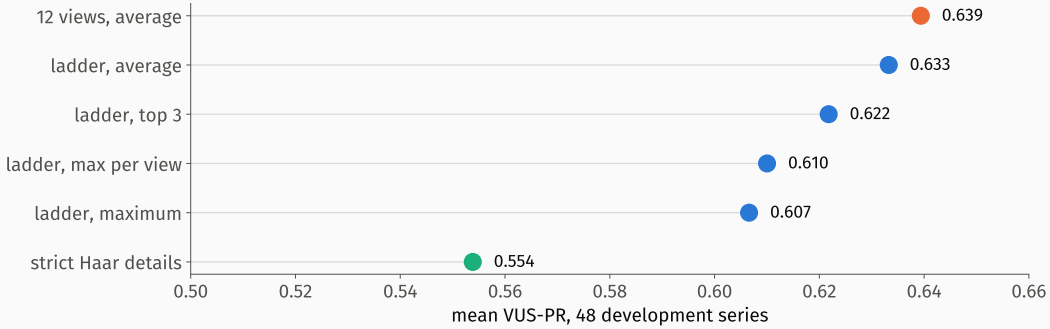
Where the ladder helps, and where it costs



Which scale for which anomaly



Combining differently does not help



What it means

scales help where they act

+0.066 on the shortest

more scales, no gain

−0.006; no combination rule fixes it

shape and L matter

−0.046 and −0.036 without them

geometry still adds

+0.008 on top of the ladder

Open question: which members to trust

noisy members, not diluted ones?

weight each member by how reliably it behaves on healthy data